

Teste Abrangente de Formulas Matematicas — Typst

1. Aritmetica Basica e Fracoes

$$\begin{aligned}a + b &= c \\ a - b &= c \\ a \times b &= c \\ a \cdot b &= c \\ \frac{a}{b} &= c \\ \frac{a + b}{c + d} &= e \\ a^2 + b^2 &= c^2 \\ \sqrt{a^2 + b^2} &= c \\ \sqrt[3]{x} &= y\end{aligned}$$

2. Simbolos Gregos

$$\begin{aligned}\alpha + \beta &= \gamma \\ \mathbf{A} + \mathbf{B} &= \mathbf{\Gamma} \\ \delta, \Delta, \varepsilon, \epsilon \\ \zeta, \eta, \theta, \vartheta, \iota, \kappa \\ \lambda, \Lambda, \mu, \nu, \xi, \Xi \\ \pi, \Pi, \rho, \varrho, \sigma, \Sigma, \varsigma \\ \tau, \upsilon, \Upsilon, \varphi, \phi, \Phi \\ \chi, \psi, \Psi, \omega, \Omega\end{aligned}$$

3. Relacoes e Operadores Logicos

$$\begin{aligned}a &< b \\ a &> b \\ a &\leq b \\ a &\geq b \\ a &\neq b \\ a &\approx b \\ a &\equiv b \\ a &\propto b \\ a &\in B \\ a &\subset B \\ a &\subseteq B \\ a &\supset B \\ a &\supseteq B \\ a &\cup B \\ a &\cap B \\ \varnothing \\ \forall x \in \mathbb{R} \\ \exists y \in \mathbb{N} \\ \neg(a = b) \\ a \wedge b \\ a \vee b \\ a \Rightarrow b \\ a \Leftrightarrow b\end{aligned}$$

4. Calculo e Analise

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}f(x) &= f'(x) \\ \frac{\mathrm{d}^2y}{\mathrm{d}x^2} \\ \int_a^b f(x) \mathrm{d}x \\ \int_0^\infty e^{-x} \mathrm{d}x &= 1 \\ \oint_C f(z) \mathrm{d}z = 2\pi i \\ \sum_{k=1}^n k = \frac{n(n+1)}{2} \\ \prod_{k=1}^n k = n! \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1 \\ \lim_{x \rightarrow \infty} \frac{1}{x} &= 0 \\ \frac{\partial}{\partial x}f(x, y) \\ \nabla f(x, y, z)\end{aligned}$$

5. Algebra Linear

$$\begin{aligned}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \\ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\ \det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ A^T \\ A^{-1} \\ A^\dagger \\ \mathrm{tr}(A) \\ \mathrm{rank}(A) \\ \dim(V) \\ \ker(A) \\ \mathrm{im}(A) \\ \langle u, v \rangle \\ \|v\| \\ |x| \\ \lfloor x \rfloor \\ \lceil x \rceil\end{aligned}$$

6. Funcoes Especiais

$$\begin{aligned}\sin x, \cos x, \tan x \\ \arcsin x, \arccos x, \arctan x \\ \sinh x, \cosh x, \tanh x \\ \log_a x, \ln x, \exp x \\ \max(a, \; b), \min(a, \; b) \\ \gcd(a, \; b), \mathrm{lcm}(a, \; b) \\ \lfloor x \rfloor, \lceil x \rceil, \lfloor x \rceil \\ \mathrm{sgn}(x) \\ \mathrm{erf}(x) \\ \mathrm{Res}(f(z), z = z_0)\end{aligned}$$

7. Conjuntos e Teoria dos Numeros

$$\begin{aligned}\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \\ \{x \in \mathbb{R} \mid x > 0\} \\ \mathcal{P}(A) \\ \aleph_0 \\ \beth_1 \\ \binom{n}{k} \\ n! = n \cdot (n-1) \cdot \ldots \cdot 1 \\ \binom{n}{k} = \frac{n!}{k!(n-k)!}\end{aligned}$$

8. Alinhamento de Equacoes

$$\sum_{k=0}^n k = 1 + \ldots + n \setminus = \frac{n(n+1)}{2} \setminus = \frac{n^2}{2} + \frac{n}{2}$$

$$\frac{3x+y}{7} = 9 \text{ dado } \setminus 3x+y = 63 \text{ multiplique por } 7 \setminus 3x = 63 - y \text{ subtraia } y \setminus x = 21 - \frac{y}{3} \text{ divida por } 3$$

9. Funcoes por Partes e Casos

$$\begin{aligned}f(x) &= \begin{cases} 0 & \text{se } x < 0 \\ x^2 & \text{se } 0 \leq x < 1 \\ 1 & \text{se } x \geq 1 \end{cases} \\ | \; x \; | &= \begin{cases} x & \text{se } x \geq 0 \\ -x & \text{se } x < 0 \end{cases}\end{aligned}$$

10. Decoradores e Acentos

$$\begin{aligned}\hat{x} \\ \tilde{x} \\ |x| \\ (x) \\ \dot{x} \\ \dot{\hat{x}} \\ \overbrace{a+b+c}^{\text{soma}} \\ \underbrace{a+b+c} \\ \frac{a+b+c}{a+b+c} \\ \overline{a+b} \\ a\!\!\!\text{--}\!\!\!b \\ \overline{a+b}\end{aligned}$$

11. Fisica e Quimica

$$\begin{aligned}E &= mc^2 \\ F &= G \frac{m_1 m_2}{r^2} \\ p &= mv \\ L &= r \times p \\ H\psi &= E\psi \\ \nabla \cdot E &= \frac{\rho}{\varepsilon_0} \\ \nabla \times B &= \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t} \\ \mathrm{d}S &\geq \frac{\delta q}{T} \\ pV &= nRT \\ \lambda &= \frac{h}{p}\end{aligned}$$

12. Probabilidade e Estatistica

$$\begin{aligned}P(A \mid B) &= \frac{P(B \mid A)P(A)}{P(B)} \\ E[X] &= \sum_i x_i P(X = x_i) \\ \mathrm{Var}(X) &= E[X^2] - (E[X])^2 \\ \sigma &= \sqrt{\mathrm{Var}(X)} \\ X &\sim N(\mu, \sigma^2) \\ \Phi(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} \mathrm{d}t \\ \hat{\mu} &= \frac{1}{n} \sum_{i=1}^n X_i\end{aligned}$$

13. Geometria e Topologia

$$\begin{aligned}\pi r^2 \\ 2\pi r \\ \frac{4}{3}\pi r^3 \\ d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \\ d(x, y) = \max_i \; | \; x_i - y_i \; | \\ d(x, y) = \sum_{i=1}^n | \; x_i - y_i \; | \\ \chi(M) = V - E + F\end{aligned}$$

14. Numeros Complexos

$$\begin{aligned}i^2 &= -1 \\ z &= a + bi \\ | \; z \; | &= \sqrt{a^2 + b^2} \\ z^* &= a - bi \\ e^{i\theta} &= \cos \theta + i \sin \theta \\ e^{i\pi} + 1 &= 0\end{aligned}$$

15. Series e Transformadas

$$\begin{aligned}f(x) &= \frac{a_0}{2} + \sum_{n=1}^\infty (a_n \cos(nx) + b_n \sin(nx)) \\ F(\omega) &= \int_{-\infty}^\infty f(t) e^{-i\omega t} \mathrm{d}t \\ f(t) &= \frac{1}{2\pi} \int_{-\infty}^\infty F(\omega) e^{i\omega t} \mathrm{d}\omega \\ \sum_{n=0}^\infty x^n &= \frac{1}{1-x} \text{ para } | \; x \; | < 1\end{aligned}$$

16. Fontes e Estilos Especiais

$$\begin{aligned}\mathbb{R} \\ \mathcal{L} \\ \mathfrak{O} \\ \code \\ \text{text} \\ \text{sans} \\ \mathrm{x} \\ \mathbf{x} \\ \mathbf{x}\end{aligned}$$

17. Subscritos e Sobrescritos Complexos

$$\begin{aligned}x_{i,j}^{k,l} \\ \sum_{i=1}^n \sum_{j=1}^m a_{ij} \\ \int_V f(x) \mathrm{d}V \\ \bigcup_{i \in I} A_i \\ \bigcap_{i \in I} A_i \\ \infty \\ \aleph \\ \beth \\ \frac{1}{\frac{1}{1+\frac{1}{x}}}\end{aligned}$$

18. Expressoes Aninhadas

$$\begin{aligned}\frac{1}{1 + \frac{1}{1 + \frac{1}{x}}} \\ \sqrt{a + \sqrt{b + \sqrt{c}}} \\ \frac{e^{x^2+y^2}}{x^2+y^2} \\ \frac{\partial^2 f}{\partial x \partial y} \\ \overbrace{\frac{a+b}{c}}^{\text{top}} \\ \underbrace{\frac{a+b}{c}}_{\text{bottom}}\end{aligned}$$

19. Espacamento e Tamanhos

$$\begin{aligned}a \; b \\ a \; \! b \\ a \; \; b \\ a \; \; \; b \\ a \; \; \; \; b \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2}\end{aligned}$$

20. Referencias Cruzadas e Numeracao

$$\begin{aligned}E &= mc^2 \; (1) \\ F &= ma \; (2)\end{aligned}$$

Referindo-se as equacoes (1) e (2).

21. Matrizes Avancadas

$$\begin{aligned}\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \right\} \; (3) \\ \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{pmatrix} \; (4) \\ \begin{pmatrix} 1 & 2 & \cdots & n \\ 2 & 3 & \cdots & n+1 \\ \vdots & \vdots & \ddots & \vdots \\ n & n+1 & \cdots & 2n-1 \end{pmatrix} \; (5)\end{aligned}$$

22. Delimitadores Escalaveis

$$\begin{aligned}\left(\frac{a}{b}\right) \; (6) \\ \left[\frac{a}{b}\right] \; (7) \\ \left\{\frac{a}{b}\right\} \; (8) \\ \left|\frac{a}{b}\right| \; (9) \\ \left\lfloor\frac{a}{b}\right\rfloor \; (10) \\ \left\lceil\frac{a}{b}\right\rceil \; (11) \\ \left\langle\frac{a}{b}\right\rangle \; (12) \\ \left|\frac{a}{b}\right| \; (13)\end{aligned}$$

23. Operadores Personalizados

$$\begin{aligned}\mathrm{Res}(f(z), \; z = z_0) &= \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} \; \mathrm{d}z \; (14) \\ \mathrm{Var}(X) &= E\left[(X - \mu)^2\right] \; (15) \\ \mathrm{Cov}(X, \; Y) &= E[(X - \mu_X)(Y - \mu_Y)] \; (16)\end{aligned}$$

24. Equacoes com Texto

$$\begin{aligned}Q &= \rho Av + \text{time offset} \; (17) \\ \mathcal{A} &:= \{x \in \mathbb{R} \mid x \text{ is natural}\} \; (18) \\ f(x) &= x^2 \text{ for all } x \in \mathbb{R} \; (19)\end{aligned}$$

25. Analise Complexa

$$\begin{aligned}\oint_{|z|=R} f(z) \mathrm{d}z = 2\pi i \sum_{k=1}^n \mathrm{Res} \left(f, a_k \right) \; (20) \\ f(z) = \sum_{n=0}^\infty \frac{f^{(n)}(a)}{n!} (z-a)^n \; (21) \\ \mathrm{Li}_2(z) = \sum_{k=1}^\infty \frac{z^k}{k^2} \; (22) \\ \Gamma(z) = \int_0^\infty t^{z-1} e^{-t} \mathrm{d}t \; (23) \\ \zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} \; (24)\end{aligned}$$

26. Mecanica Quantica

$$\begin{aligned}[\hat{x}, \hat{p}] = i \; \hbar \; (25) \\ i \; \hbar \; \frac{\partial \Psi}{\partial t} = \hat{H} \Psi \; (26) \\ \Psi(x, \; t) = \sum_n c_n \psi_{n(x)} e^{-iE_n \frac{t}{\hbar}} \; (27) \\ \langle \varphi | | \psi \rangle = \int \varphi^*(x) \psi(x) \mathrm{d}x \; (28) \\ \langle A^{\cdot} \rangle = \langle \psi | \hat{A} | \psi \rangle \; (29)\end{aligned}$$

27. Relatividade

$$\mathrm{d}s^2 = -c^2 \mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \; (30)$$

$$\Gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \; (31)$$

$$E^2 = (pc)^2 + (mc^2)^2 \; (32)$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \; (33)$$

28. Teoria dos Grafos

$$\chi(G) \geq \omega(G) \; (34)$$

$$\alpha(G) + \tau(G) = | \; V(G) \; | \; (35)$$

$$\sum_{v \in V(G)} \deg(v) = 2 \mid E(G) \mid \; (36)$$

$$K_n, C_n, P_n, Q_n \; (37)$$

29. Teoria da Informacao

$$H(X) = - \sum_{x \in X} P(x) \log_2 P(x) \; (38)$$

$$I(X; Y) = H(X) - H(X \mid Y) \; (39)$$

$$D_{KL}(P \mid Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \; (40)$$

$$C = \max_{P_X} I(X; Y) \; (41)$$

30. Otimizacao

$$\min_{x \in \mathbb{R}^n} f(x) \text{ sujeito a } g_{i(x)} \leq 0, h_{j(x)} = 0 \; (42)$$

$$L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_{i(x)} + \sum_j \mu_j h_{j(x)} \; (43)$$

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla g_{i(x^*)} + \sum_j \mu_j^* \nabla h_{j(x^*)} = 0 \; (44)$$